

THE MYSTERY OF THE NEGATIVE SQUARE

An Introduction to Complex Numbers • Level 1: The Question

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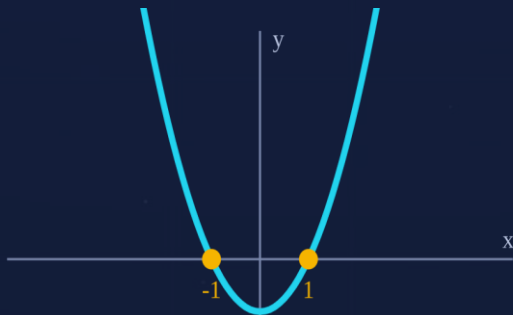
Concept: Squaring any real number never gives a negative result. So does a solution exist for $x^2 = -1$?

"Can a real number square to -1...?"



EQUATION 1

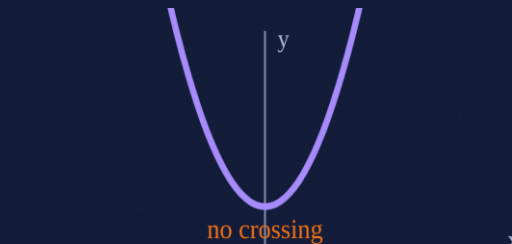
$$x^2 - 1 = 0$$



$x = \pm 1$ — real roots exist

EQUATION 2

$$x^2 + 1 = 0$$



$x = \pm\sqrt{-1} = ?$

⚠ WARNING SIGNAL

The graph of $f(x) = x^2 + 1$ never touches the x-axis — it has no real x-intercepts, so no real number solves the equation.

🌟 KEY TAKEAWAY

Real numbers alone cannot solve $x^2 = -1$ — mathematics needs a brand-new kind of number.

MEET THE IMAGINARY UNIT

Level 2: The Discovery

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$$i = \sqrt{-1}$$

the imaginary unit

$$i^2 = -1$$

the defining rule of every power of i



CONCEPT

Because equation 2 has no real solution, mathematicians created a new number whose square *is* allowed to be negative.

Formula card: $x = \pm\sqrt{-1}$ is written as $x = \pm i$

✦ **Key takeaway:** Every power of i can now be built from this one fact — $i^2 = -1$ unlocks the entire pattern.

REAL vs IMAGINARY

Real: $x^2 \geq 0$ always

Imaginary: $i^2 = -1$

This single new rule lets us find a square root for any negative number.

THE POWER CYCLE OF i

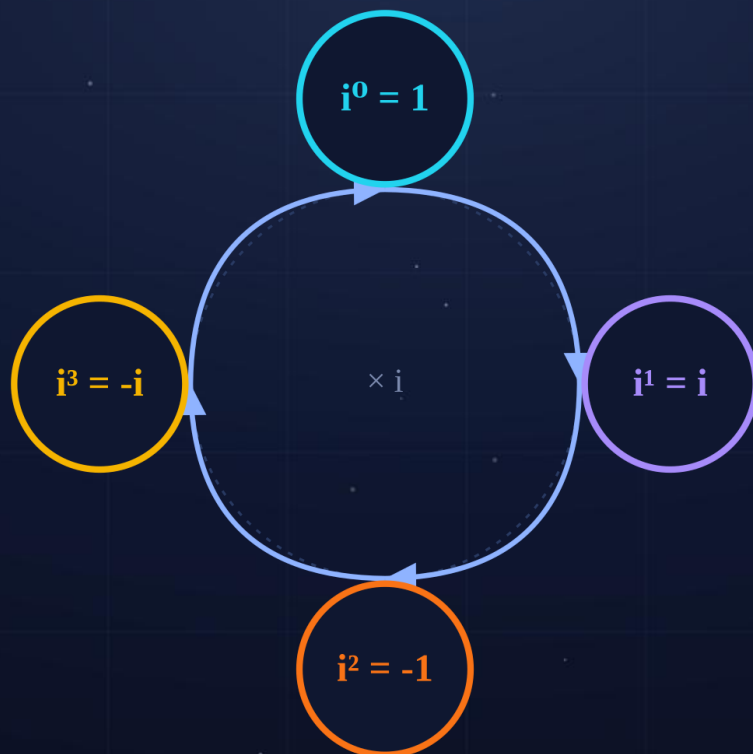
Level 3: Rules + Worked Examples

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Rule: $i^n = i^k$ where $k = n \bmod 4$

WORKED EXAMPLES

Simplify i^7

$$i^7 = i^{4+3} = (i^4) \cdot i^3 = 1 \cdot i^3 = -i$$

Simplify i^{1729}

$$1729 \div 4 = 432 \text{ rem } 1 \rightarrow i^{1729} = (i^4)^{432} \cdot i^1 = i$$

Simplify $i^{-1924} + i^{2018}$

$$\text{both exponents} \equiv 0 \pmod{4} \rightarrow i^0 + i^0 = 1 + 1 = 2$$

FOUR BASIC VALUES

$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = -i$$

✓ PATTERN

Powers of i repeat every 4 steps: 1, i , -1 , $-i$, then back to 1. Divide the exponent by 4 — only the remainder matters.

MASTERING THE RESULT

Level 4: Key Result, Caution & Practice

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IMPORTANT RESULT

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$$

The sum of any four consecutive powers of i is always zero, for every integer n .

CAUTION

$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$ only holds if at least one of a, b is non-negative.

Counter-check: $6 = \sqrt{36} = \sqrt{(-4)(-9)} \neq \sqrt{-4} \cdot \sqrt{-9} = (2i)(3i) = -6$

QUICK PRACTICE

Q1 $i^{1947} + i^{1950}$

Q2 $i^{1948} - i^{1869}$

Q3 $\sum i^n$ for $n = 1$ to 12



CONCEPT UNLOCKED